

Report – RSA for Model to Brain Analysis

Introduction

This study looks at how similar artificial neural networks are to the human brain when they are processing the same sounds. Specifically, we saw how different models represent sounds compared to how the superior temporal gyrus (STG), which is a part of the brain that processes sound represents them.

Data and Models

For the brain data we used the Santoro dataset, which contains fMRI responses from the STG to 288 natural sounds across ten categories.

For the models we looked at 3 different models, which are waveform, uninspired and inspired. Each one of them was tested untrained, built from scratch with no learning and trained, which was loaded from five saved checkpoints for each model. We also included YAMNet, a pretrained audio model, as an extra comparison point.

Methods

- Building RDMs

For the brain data, we computed a Representational Dissimilarity Matrix (RDM) , which is a table showing how different each pair of sounds is using correlation distance. We did the same thing for each layer of each model, using the activations it produced for the 288 sounds. This produced one RDM for each model's layer, so they can be compared directly.

- Spearman Correlation

To measure how well a model layer matches the brain, we used Spearman correlation between the model's RDM and the brain's RDM. Higher correlation means the model is more similar to how the brain sorts sounds. After we averaged each trained model's five runs for comparisons.

- Comparing Method

We took the average trained score for each layer and subtracted the untrained score for the same layer. Here, positive values mean that training helped and negative values mean the opposite. Then, we compared the average scores of the 3 trained models. We looked at the 2 plots to see how alignment changes if you go deeper into the model.

- Statistics

Because each trained model has 5 runs, we calculated the average over the runs and looked at the standard deviations and means for each layer.

- t-SNE

We used t-SNE to transform the activations of the trained inspired model to 2D so that plotting the 288 sounds as points could be possible and each category would have its color. We did this for 3 convolutional layers and one recurrent layer.

Results

- RSA Summary

Model	Condition	Spearman_r
Inspired	Trained	0.067118
Inspired	Untrained	0.086595
Uninspired	Trained	0.037480
Uninspired	Untrained	0.047634
Waveform	Trained	0.006758
Waveform	Untrained	-0.010082
YAMNet	Pretrained	0.098000

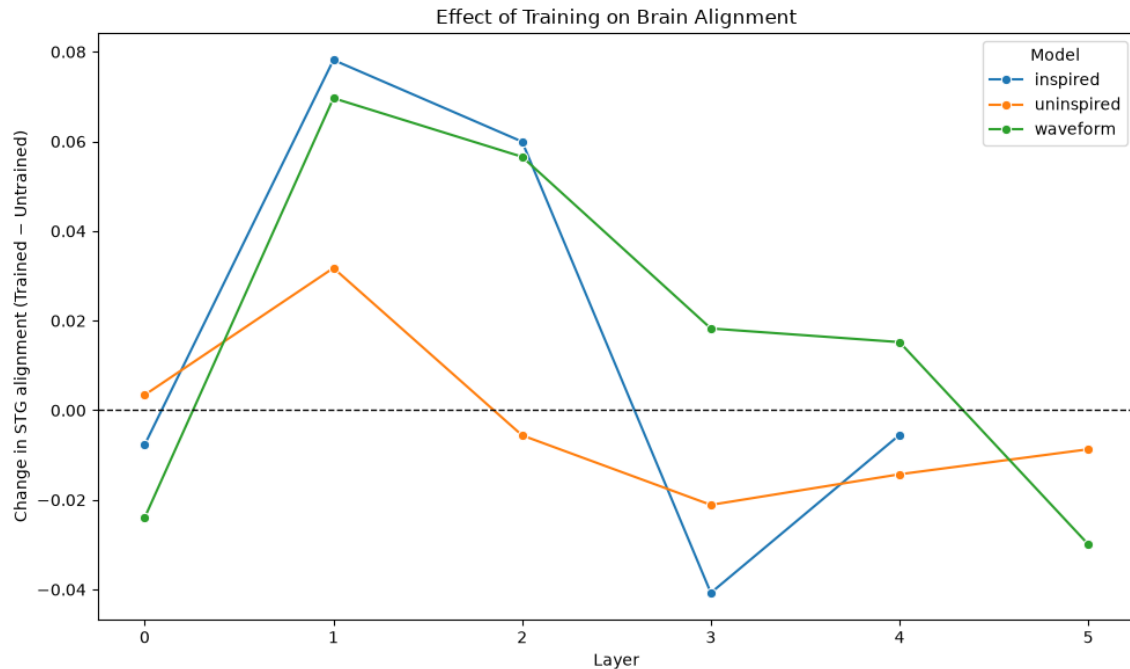
Although the differences are small, the untrained models for inspired and untrained have a higher spearman value than the trained versions.

Inspired training gives a small boost in the middle convolutional layers (layer 1 and 2), but has a negative impact in the deeper layers.

Uninspired training helps a tiny bit in the earliest layers, then has a negative impact from layer 2 onwards.

Waveform training helps in the middle convolutional layers, but not at the start or at the classifier.

- Effect of Training on Brain Alignment



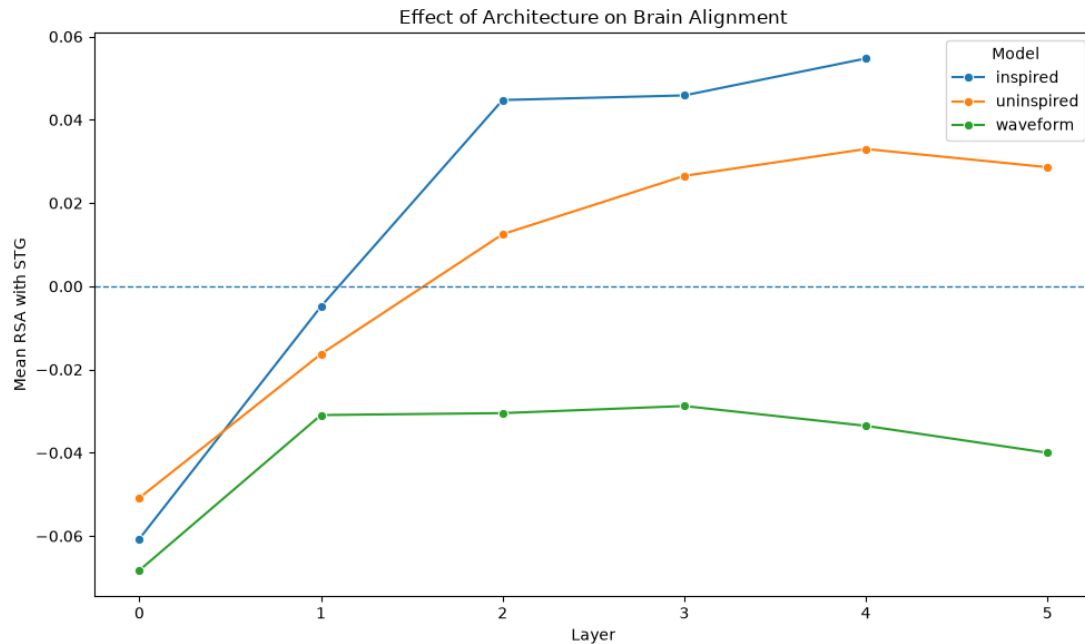
Given the change in the three model's RSA scores (mean of trained - spearman correlation of untrained) we can see that positive value mean the training model is more similar to the STG while negative mean it's less similar:

Although the change is small, for the inspired model RSA remains positive until the third layer and after that it stays in the negative part.

For uninspired model RSA change remains in positive until second layer and until layer 4 for waveform, again, the change is relatively low and after that they remain negative with relatively low change in RSA.

Additionally, all 3 models increase in change in layer 1 while decreasing afterward with inspired having the sharpest decline between layer 2 and 3. While increase is limited, inspired and uninspired do see an increase after layer 3.

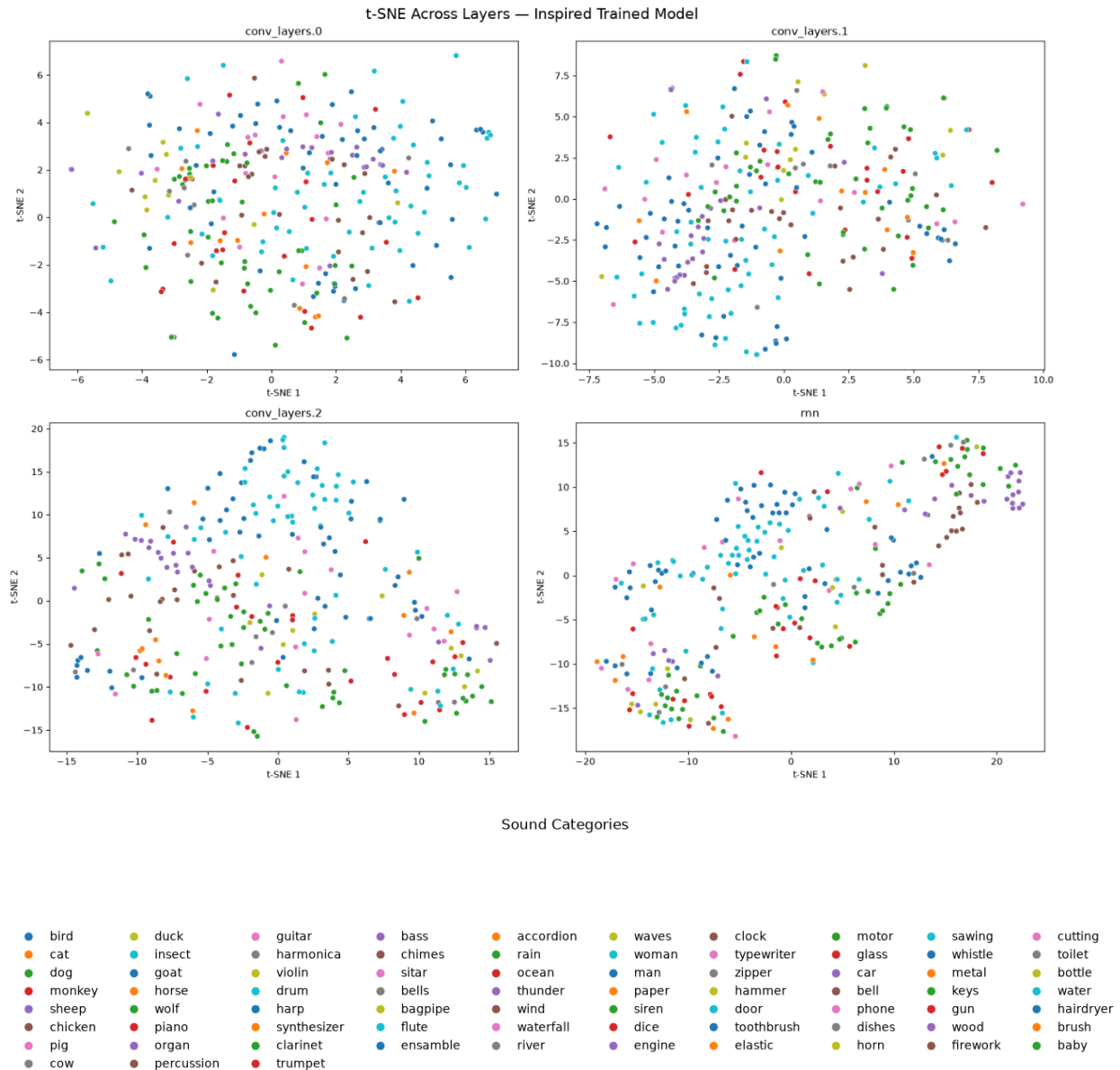
- Effect of Type of Model Choice on Brain Alignment



While all 3 models, inspired, un-inspired and waveform, remained in the negative area before layer 1, both inspired and un-inspired models rose to the positive RSA area after layer 1 (with waveform remaining in negative area).

The inspired model showed the highest brain alignment while un-inspired showed less with waveform remaining negatively aligned. Inspired model also shows greater increase with un-inspired slowing down its increase and eventually dipping slightly and waveform almost flatting before decreasing in the deeper layers again.

- t-SNE Results



The t-SNE visualization of the trained model shows 288 sound stimuli on a 2D graph. Each point is a stimulus and the color is a category. In the earlier layers there is a relatively large amount of overlap, with deeper layers showing less overlap (overlap still remains but becomes less as layers get deeper) and showing more distinct groups of stimuli. We can see that as layers get deeper, more structured distribution of the representations/organization becomes apparent with several groups becoming more spatially apparent compared to former layers.

Discussion

- Untrained vs Trained

The training affects very little the results of the Spearman values. Although, we expected trained models to perform better, untrained models in this case matched or are even better than the trained ones and the layer-level changes were small and went in both directions. Additionally, based on the Effect of Training on Brain Alignment, we can conclude, observing the low value of change, that training affects the alignment modestly and is layer dependent rather than consistently improving or making the alignment worse.

- Architecture Choice Consequences

Compared to training, the choice of the architecture model matters more. The Effect of Type of Model Choice on Brain Alignment results suggest that different architectures are associated with different varieties of alignment per layer and inspired alignment shows a relatively greater alignment across layers. The inspired model's design seem more suited to processing auditory data than the other architectures.

- Depth's role in Alignment

Alignment improves with depth for the models that can reach positive alignment. Based on the Effect of Type of Model Choice on Brain Alignment, we can see that uninspired and inspired models generally show increasing alignment in deeper layers, while the waveform model remains negatively aligned across all 5 layers. This suggests that both inspired and uninspired models show a more similar representation in deeper layers to the structure in STG.

- t-SNE

The t-SNE visualization helps us understand why mid-depth layers align best. That is because they're where category structure first emerges without being fully split apart. Based on the t-SNE results we can see that as layers get deeper, more structured distribution of the representations/organization becomes apparent with several groups becoming more spatially apparent compared to former layers.

Future Work

For the statistics part, an improvement would be to treat each run as its own data point and run paired permutations tests comparing trained and untrained alignment for each layer.

For modelling, adding more biological detail always helps. Adding, for example, unsupervised pretraining on a larger sound dataset with fine-tuning could also have a closer result to the brain than training on a small labeled dataset.

Conclusion

In this study we compared untrained, trained, and pretrained models against STG brain responses to natural sounds. Training changed alignment modestly and inconsistently across layers, rather than making models more brain like. Architecture mattered more, which was shown when the inspired model beat uninspired and waveform. Depth helped up to middle-to-deep layers, but only for models that were already capturing brain like structure. The t-SNE visualization supports these findings by showing that mid-depth layers are where category structure emerges. Future work combines proper statistical testing with richer biological detail in the models.